Optimal design in pharmacokinetic studies

A brief introduction

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What is optimal design?

In context of pharmacokinetics

- Optimal design is the discipline of planning an experiment to extract the **most information** from the **fewest resources**.

- Optimal design treats the experiment itself as a decision, choosing ***what*** to observe, ***when***, and ***from whom***, so that the resulting data answer the scientific question with maximum precision per unit of cost.

- In pharmacometrics, optimal design answers a question that precedes any modeling e.g., given a model, a budget, and a clinical reality, what choice of sampling times, doses, and subjects yields parameter estimates we can defend?

This approach is important in fields like drug development, where experiments can be costly and time-consuming.

Framework and definition

We design and choose of experiment ξ that maximizes information about an unknown parameter θ, subject to operational constraints

$$\xi^* = \arg\max_{\xi \in \Xi} \Phi(\xi) \quad \text{subject to} \quad C.$$

$$\xi = (\underbrace{x_t}_{\text{sampling times}}, \underbrace{a}_{\text{doses}}, \underbrace{N}_{\text{subjects}}, \underbrace{n_t}_{\text{samples/subject}}, \underbrace{z}_{\text{strata}})$$

Formally, a PK model $y = f(x, \theta) + \varepsilon$ with parameter and a design space B (every experiment the protocol could specify). An information measure Φ and constraints (ethical, operational and logistics)

The optimal design is the answer to a constrained optimization: of every experiment we “could” run, which carries the most information about the parameter (endpoints) without breaking the rules?

Motivating example

Pediatrics

Pediatric trials operate under constraints (**c**)

- Sparse sampling: typical protocols allow 2 to 4 blood draws per child across a dosing interval, with strict total-volume limits that scale with body weight.
- Heterogeneous populations: neonates, infants, children, and adolescents differ in clearance pathways, body composition, and protein binding. A single design may need to serve all of them.
- Extrapolation from adults: the sponsor begins with adult PK and must demonstrate, with limited data, that the pediatric exposure is comparable enough to support a labeled dose.
- Regulatory agencies (FDA, EMA, PMDA) increasingly require MIDD pediatric development plans before a study is approved.

Motivating example

Pediatrics

The design problem reduces to:

- Model: structural PK with allometric scaling on CL and V

- Design vector: $\xi = (t_1, \ldots, t_n)$ sampling times within $\mathbf{c}$

- Goal minimize the uncertainty of $\hat{\theta}$ for $\theta = (CL, V, k_a, \ldots)$.

- Constraints ($\mathbf{c}$): maximum samples per child, total blood volume per visit, clinically feasible time windows.

Where does information come from?

We have a definition: $\xi^* = \arg\max_{\xi \in B} \Phi(\xi)$ and we have a motivation: pediatric trials, sparse sampling, regulatory stakes. What we don't have yet: a way to compute the information Φ from a model and a design?

Information about θ exists only where the model output responds to changes in θ. If the prediction at x doesn't move when θ moves, no data at x can teach us about the parameter.

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f(x, y) = 5x. Does the function change with y? Does the function
depends on y?
```

That responsiveness has a name: **sensitivity**

Sensitivity

The building block of every information measure

For a model $y = f(x, \theta)$, the sensitivity of the prediction at point x to parameter θ_j is the partial derivative

$$S_j(x, \theta) = \frac{\partial f(x, \theta)}{\partial \theta_j}.$$

Where:

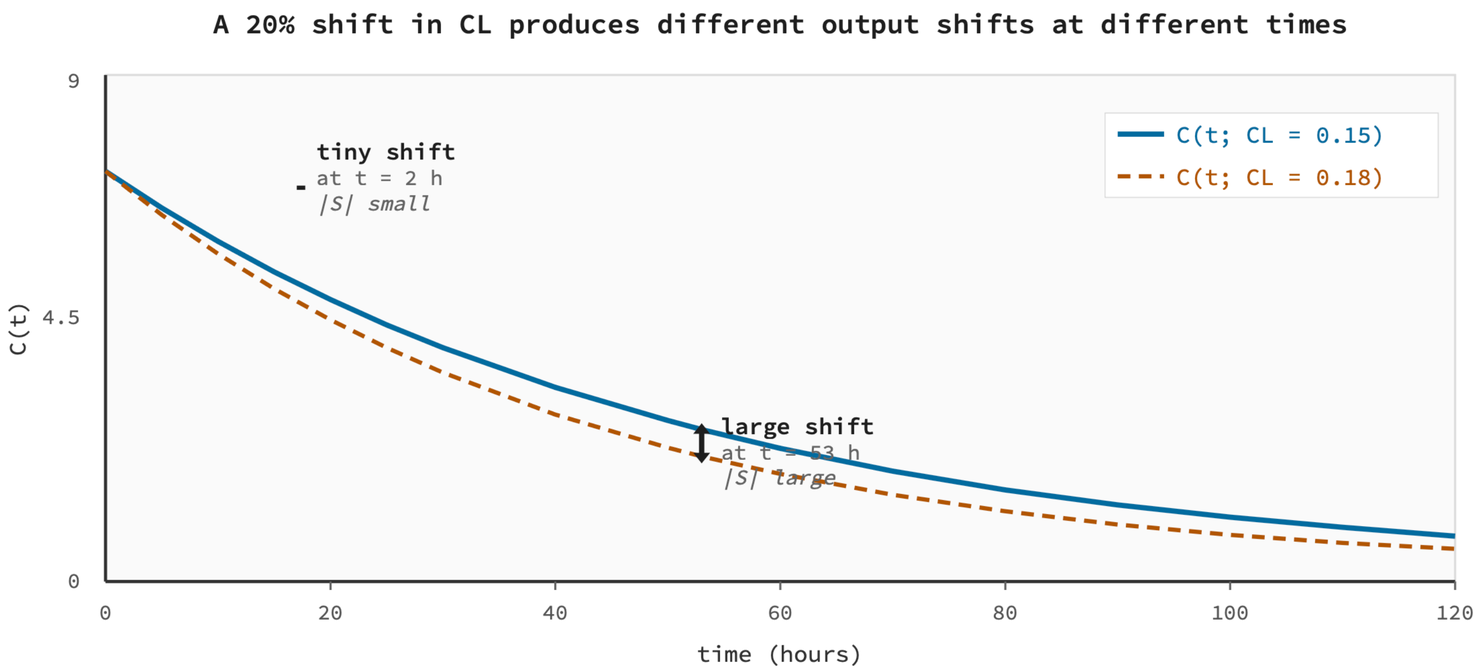
- $|S_j(x, \theta)|$ large at x → a small change in θ_j shifts the prediction noticeably; data at x is informative about θ_j.
- $|S_j(x, \theta)|$ small at x → θ_j barely affects the prediction; data at x tells us little.
- $|S_j(x, \theta)| = 0$ at x → θ_j is locally invisible at that design point.

Sensitivity

The building block of every information measure

Parameters are never measured directly and only their effects on observations are.

Sensitivities tell you, for each design point xt, how strongly each parameter influences what you would record there.



Example with the PK "hello world"

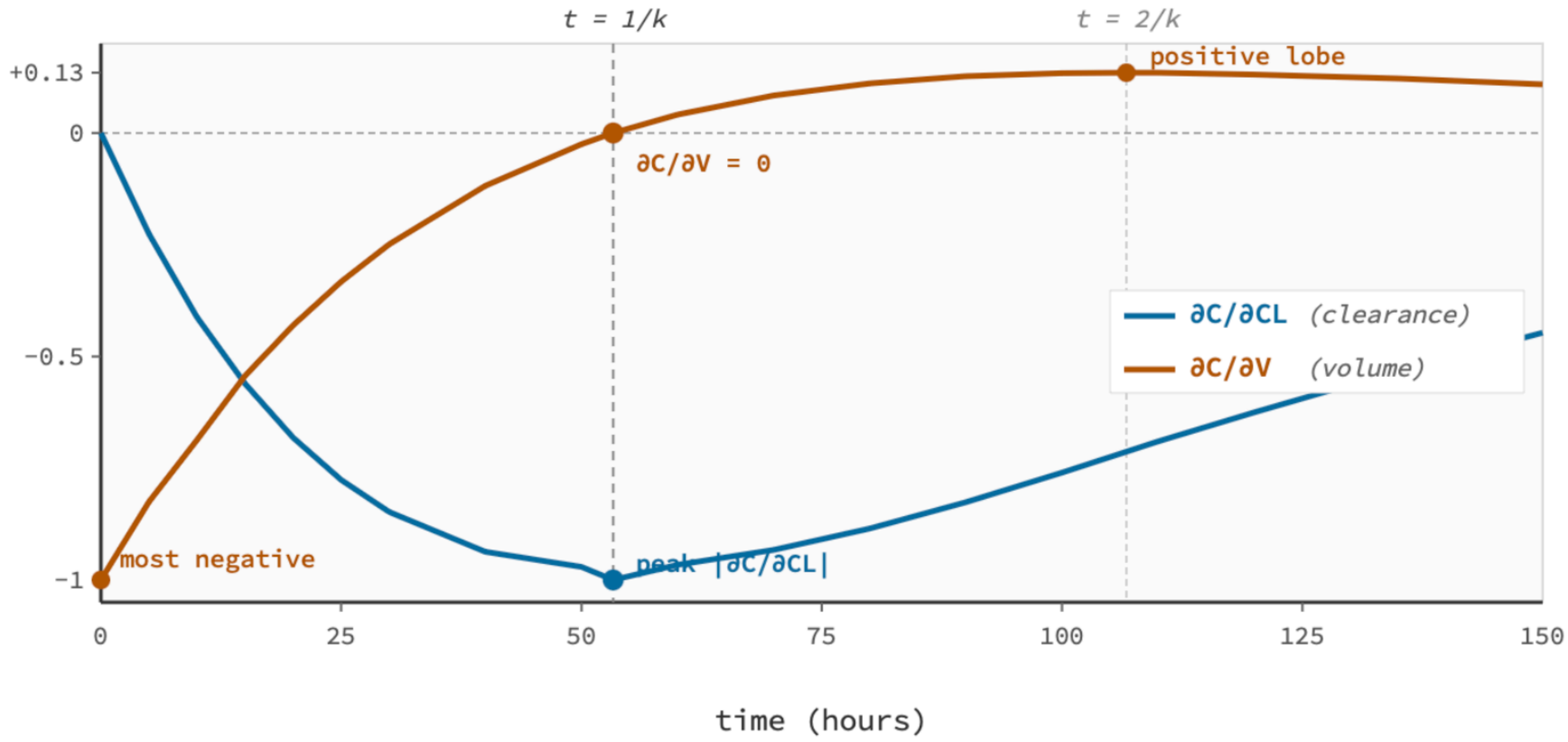
$$C(t,\theta) \;=\; \frac{D}{V}\,\exp(-k\,t), \qquad k = CL/V\,.$$

Compute the sensitivities (partial derivative)

$$\frac{\partial C}{\partial CL} \;=\; -\,\frac{D\,t}{V^2}\,\exp(-k\,t) \qquad\qquad \frac{\partial C}{\partial V} \;=\; \frac{D}{V^2}\,(k\,t-1)\,\exp(-k\,t)$$

$\partial C/\partial CL$ maximum at $t^* = 1/k$ the elimination time-constant equivalent.

$\partial C/\partial V$ crosses zero at $t^* = 1/k$, negative before, positive after.

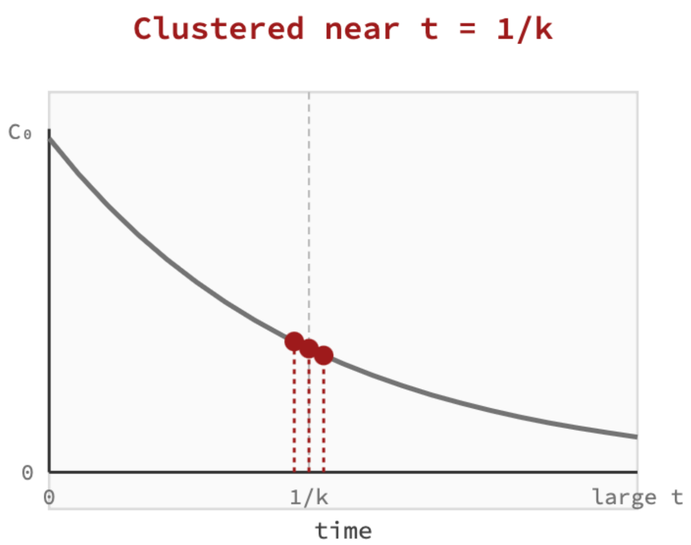


Two things to notice:

1. CL |S| is 0 at t=0, then peaks around 1/k, then decays

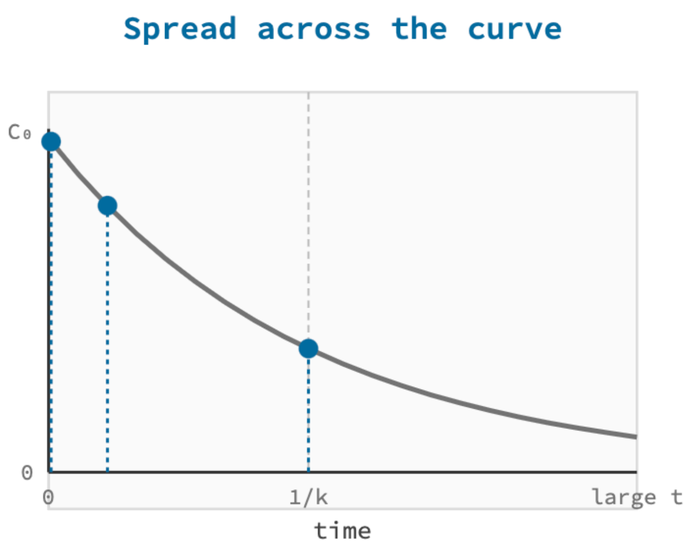
2. V |S| is large at t=0, cross 0 at 1/k and then rises to a positive lobe then slowly decays to zero.

Sign does not matter, but the magnitude does



det M = 1.45

RSE(CL) = 18% · RSE(V) = **385%**



det M = 859

RSE(CL) = 29% · RSE(V) = **10%**

Optimal design is rarely "all samples at the most informative time." It is a **spread** that balances information across parameters.

From sensitivity to information

$$S_i(\theta) \;=\; \Big(\partial f(x_i,\theta)/\partial\theta_1,\; \ldots,\; \partial f(x_i,\theta)/\partial\theta_p \Big)$$

The Fisher Information Matrix for design with residual variance is the weighted outer-product sum (fixed-effect only):

$$M(\theta,\xi) \;=\; \frac{1}{\sigma^2}\sum_{i=1}^{n} S_i(\theta)^\top S_i(\theta)\,.$$

The FIM is just an outer-product accumulator. Each sample contributes one piece. The matrix tells us how much information the whole design carries about θ

Why does the FIM matter for design? $\mathrm{cov}(\hat{\theta}) \geq \left[M_F(\theta,\xi)\right]^{-1}$

Cramér–Rao lower bound

- For any unbiased estimator $\hat{\theta}$ of θ

$$\mathrm{cov}(\hat{\theta}) \geq \left[M_F(\theta, \xi)\right]^{-1}$$

- $Cov(\theta)$ is the actual covariance matrix of your estimator
- $\left[M_F(\theta, \xi)\right]^{-1}$ is the inverse FIM (refer to the lower bound of parameter precision)

Approximately for maximum likelihood of large sample.

Back to our example

For our 1-comp IV bolus, the per-sample contribution is the outer product $M_i = \frac{1}{\sigma^2} S_i^\top S_i$

Plugin the sensitivities $$M_i = \frac{D^2 e^{-2kt_i}}{\sigma^2 V^4} \begin{pmatrix} t_i^2 & -t_i(kt_i - 1) \\ [2pt] - t_i(kt_i - 1) & (kt_i - 1)^2 \end{pmatrix}.$$

Total for all points $$M(\theta, \xi) = \sum_{i=1}^{n} M_i .$$

Continue in week15/ex1.R

Beyond the FIM: stochastic simulation and estimation

Another common method is stochastic **simulation-estimation (SSE)** and its gold standard (brutal force). It includes:

1. Simulate K datasets from the model under candidate design ξ.
2. Estimate $\hat{\theta}_k$ from each dataset
3. Evaluate RSEs for parameters

however, workflow can be considered **computationally expensive** and is not different with large N

Note on identifiability and estimability

- Structural identifiability (*PK/PD model structure*) is necessary but not sufficient. A clean model can become practically un-estimable under a poor study design.

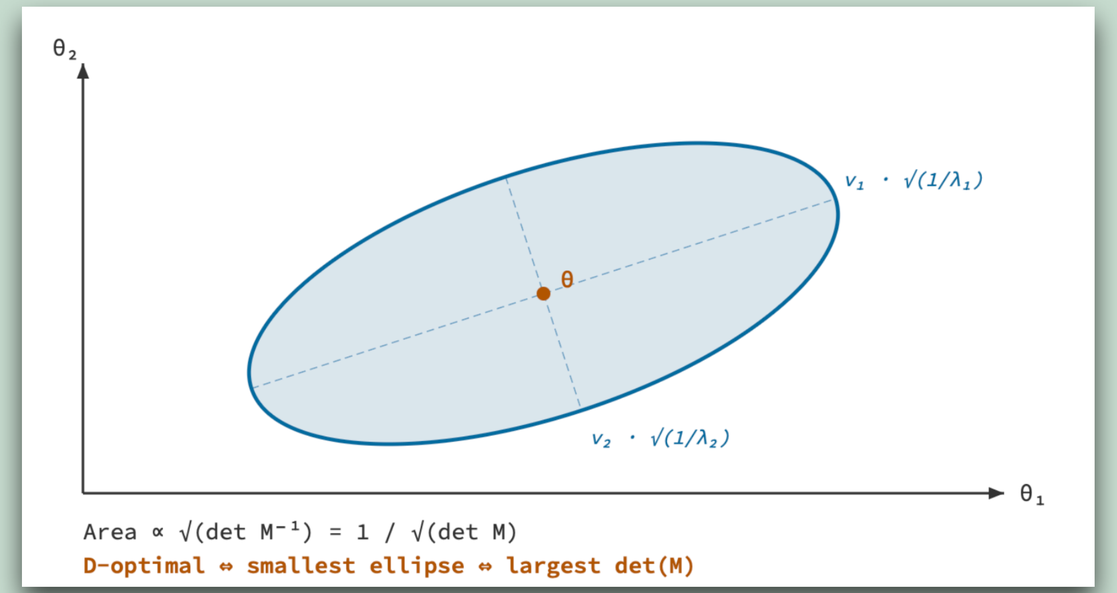
- θ is practically estimable under design ξ if the inverse of FIM exists and doesn't blow up along any direction. Both the estimate and the standard errors

We turn the matrix FIM into a single number $\Phi(M^{-1})$ The smaller Φ, the better the design and the choice of value is the **optimality criterion**

Optimality criteria

- D-optimality $\Phi_D(\xi) = -\log\det M(\theta, \xi)$

- Minimize the -log det FIM (or equivalent to maximize the det FIM)
- Minimize the volume (area) of join ellipsoid for θ
- Invariant under linear reparametrization (pleasant property)
- Default when all parameters matter

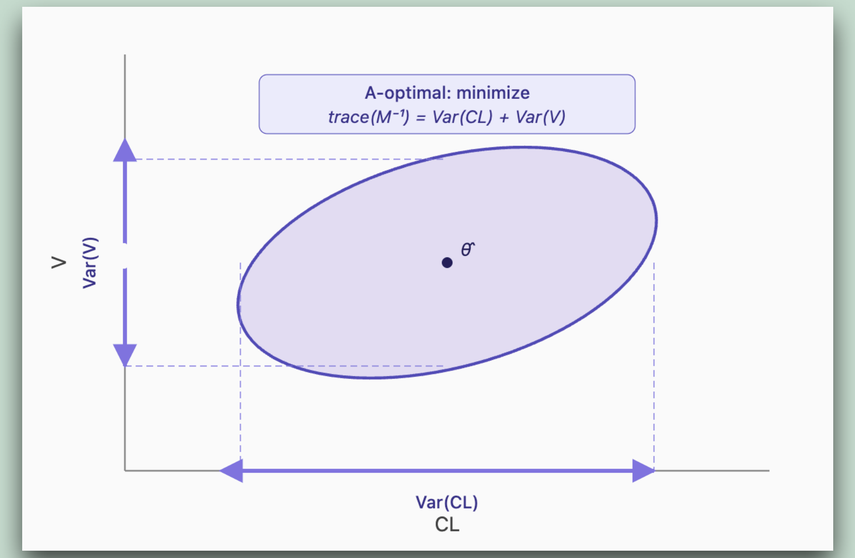


Optimality criteria

- A-optimality

$$\Phi_A(\xi) \;=\; \mathrm{tr}\, M(\theta, \xi)^{-1}$$

- Minimize the sum of parameter variance (axis-length) the average uncertainty across θ
- Sensitive to parameter units e.g., Var(CL) + Var(V),
- Can be normalized to trace of normalized variances
- Reduce overall uncertainty

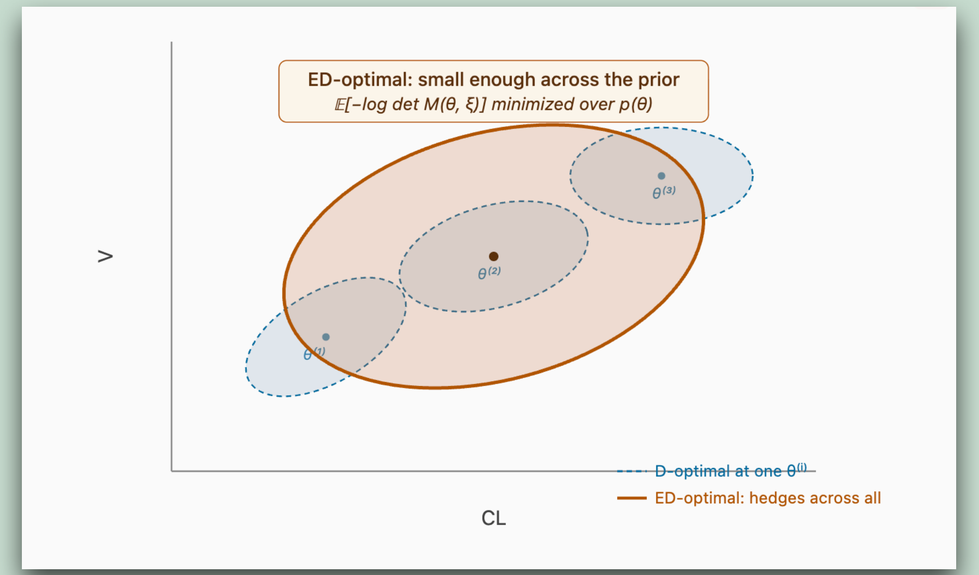


Optimality criteria

- ED-optimality (using prior knowledge)

$$\Phi_{ED}(\xi) \;=\; \mathbb{E}_{\theta\sim p(\theta)}\left[-\log\det M(\theta,\xi)\right]$$

- D-optimality requires plugging in a value for θ
- ED averages the D-criterion over a prior $p(\theta)$
- The design search against parameter uncertainty instead of optimizing over one-guess
- Common in FIH, pediatric extrapolation or in case of θ is uncertain

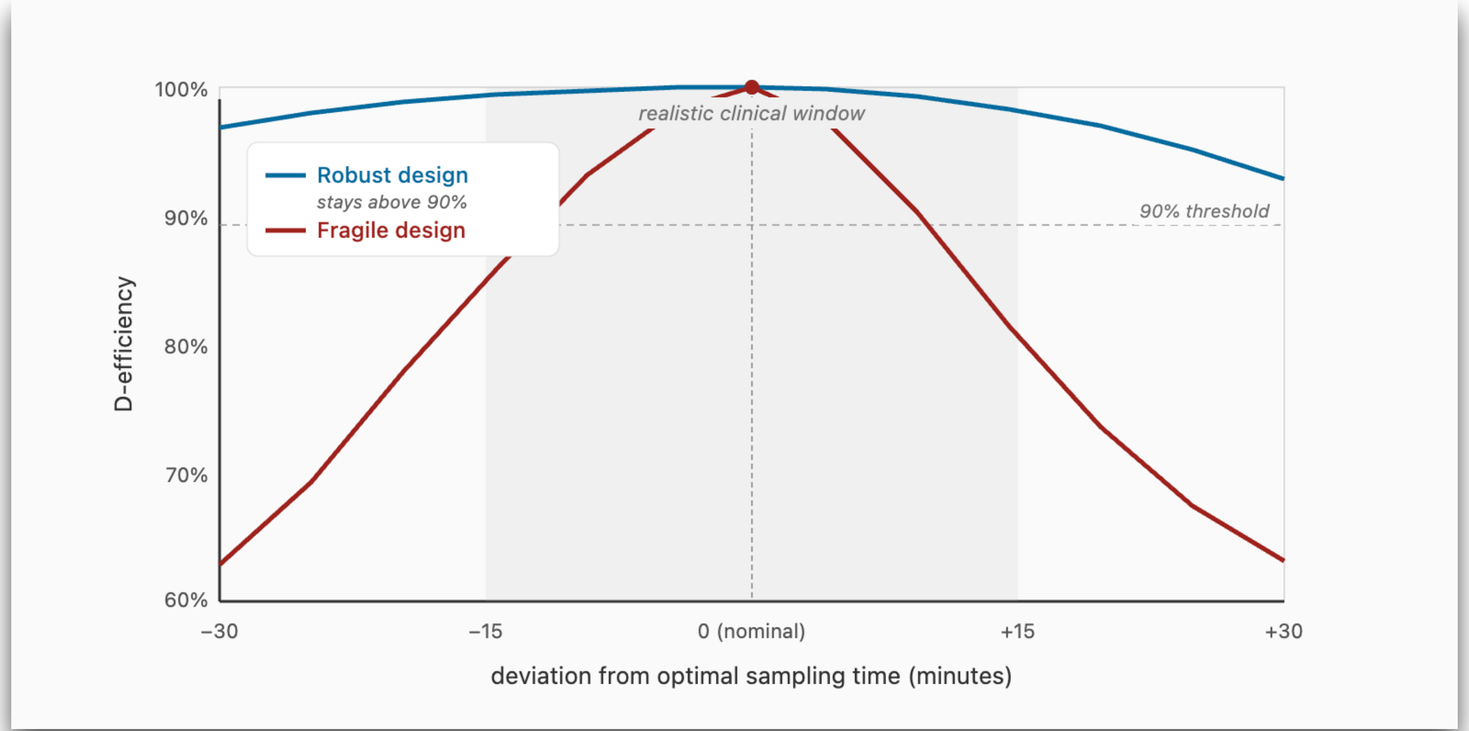


Near-optimal design - Clinical plausibility

A D-optimal design might say "sample at exactly t = 0.184 h."

how much D-efficiency do we lose if the sample lands anywhere in that window?

A robust design is one whose efficiency stays high across realistic windows; a fragile design's efficiency collapses if the sample drifts even a few minutes.



Design efficiency as a practical metric

$$Eff_D(\xi, \xi^*) = \left(\frac{det\, M(\xi)}{det\, M(\xi^*)}\right)^{\frac{1}{p}}$$

- Where ξ^* is the D-optimal design (not near-optimum) compared to the initial or reference design ξ.
- Number of parameters p in the optimized model
- Why does this number matters? If EffD is 80%, then that mean we need 1/0.8 (25% increase) fold of individuals to match the optimal design (or to recover lost information).
- Look at RSEs if you need to evaluate number of samples selected.

Are we only optimizing sample times?

NO.

Back to design matrix

$$\xi = (\underbrace{x_t}_{\text{sampling times}}, \underbrace{a}_{\text{doses}}, \underbrace{N}_{\text{subjects}}, \underbrace{n_t}_{\text{samples/subject}}, \underbrace{z}_{\text{strata}})$$

All elements in ξ can in be optimized based on defined criteria.

1. *Dose evaluation for Drug X that follow a nonlinear PK?*
2. *How many individuals needed for the trial to achieve enough power?*
3. *How many samples per group I need to have enough information about covariates*

Tools used for optimal design

The common ones. Other exists

Tool	Documentation
PopED	Common, and have a good documentation: https://andrewhooker.github.io/PopED/
PFIM	https://github.com/packagePFIM/PFIM
NONMEM	Using `$DESIGN` using FO/FOCE algorithms https://ascpt.onlinelibrary.wiley.com/doi/epdf/10.1002/psp4.12713
Pumas, OptimalDesign.jl	https://tutorials.pumas.ai/html/introduction/optimal_design.html#expected-fisher-information

Limitations

- The Cramér-Rao bound $\mathrm{cov}(\hat{\theta}) \geq M^{-1}$ is tight only asymptotically, for large N with well-behaved likelihoods. In small samples, the FIM underestimates true parameter uncertainty, sometimes severely.

- The FIM assumes the estimator is unbiased. FOCE and SAEM are not guaranteed unbiased in finite samples with nonlinear mixed-effects models.

- Optimal design is only optimal for the assumed structural model. The optimal times with severe model misspecification may be uninformative or even misleading.

- The determinant is dominated by the largest eigenvalue of M. A design can have high det M while leaving one parameter direction almost unidentifiable

- FIM ignores BLQ information

Let’s do some examples